Data 250 Practice Exam 2

1. Does there exist a simple graph with 7 vertices each with degree 3? Why or why not?

2. Define the Ackermann function A(m,n) with this recursive definition and calculate A(2,2)

$$A(m, n) = \begin{cases} 2n & \text{if } m == 0 \\ 0 & \text{if } m \geq 1 \text{ and } n == 0 \\ 2 & \text{if } m \geq 1 \text{ and } n == 1 \\ A(m-1,\ A(m, n-1)) & \text{if } m \geq 1 \text{ and } n \geq 2 \end{cases}$$

3. Give a recursive definition for x_n = n-th positive even number

4. Show if a divides b and b divides a for positive integers a,b, then it follows that a = b

5. Prove or disprove: if a,b,c, m are integers ≥ 2, then
a c = bc mod m implies that a = b mod m

6. Two positive integers x and y are relatively prime if gcd(x,y) = 1.
Find all pairs in this list which are relatively prime.

Out[ ]= {80, 11, 95, 7, 38}

7. Find two sets A and B such that A ∈ B and A ⊂ B

8. Show that if $2^n + 1$ is prime where n > 0, then n must be a power of 2.